

An Optimized Geographic Routing Scheme in SDN-Enabled 5G UAV Networks

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Abstract—Unmanned Aerial Vehicle (UAV) networks are essential for next-generation communication systems, enabling collaborative swarm operations for applications such as disaster response, surveillance, search, and rescue. However, the dynamic nature of UAV swarms, characterized by high mobility and frequent topology changes, poses significant challenges for efficient and reliable communication. Existing routing protocols, including traditional geographic routing protocols, struggle to maintain optimal performance in such a dynamic environment. To address these issues, we propose in this paper a novel mechanism called GeoSDN-UAV, an SDN-assisted geographic unicast routing protocol, that integrates Greedy Perimeter Stateless Routing (GPSR) with a Software-Defined Networking (SDN) architecture and 5G-enabled ground stations. Using SDN's global network view, the controller dynamically optimizes routes based on real-time mobility of UAVs and link stability, ensuring resilient and efficient data delivery. GeoSDN-UAV is based on 5G ground stations to provide ultra-low latency communication, allowing instantaneous topology updates and routing adjustments when the UAV disconnections occur. The evaluation shows that GeoSDN-UAV significantly optimizes the performance of the UAV network in terms of packet delivery ratio, end-to-end delay, and routing overhead, particularly in high-mobility the UAV environments. Our proposal establishes GeoSDN-UAV as a scalable and reliable solution for the UAV in next-generation 5G and beyond networks.

Index Terms—UAVs; Geographic Routing Protocol; SDN; 5G and Beyond Networks;

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have become vital in civilian and military applications such as disaster response, surveillance, and control. the UAVs fly like autonomous systems to collaborate between them for different purposes. The UAV applications, require multi-hop communication between the UAVs [1]. Thus, strong communication protocols are needed to maintain stable connections and guarantee reliable data transfer, improving scalability and supporting mobility. However, achieving the full potential of UAVs depends on a well-structured network architecture that ensures seamless communication [2].

Three different schemes of communications are possible in the UAV networks: (i) UAV-to-UAV communication for data sharing and connectivity. (ii) UAV-to-Ground communication for command and control via ground control stations (GCS). Integrating ground stations with 5G is a promising technology

[3] that can provide high-speed, low-latency communication for enhanced real-time data exchange and mission coordination. Despite these advancements, an efficient routing protocol is required due to high mobility, frequent topology changes, and intermittent connectivity.

Traditional routing protocols designed for Mobile Ad Hoc Networks (MANETs) and Vehicular Ad Hoc Networks (VANETs) [4] [5] are not suitable for the UAVs because they depend on stable connections and fixed paths. The 3D movement of UAV often leads to broken links, high packet loss, excessive retransmissions and eventually increased delay, which makes routing in the UAV networks even more challenging. The Geographic Unicast routing protocol, which sends data to specific geographic positions [6], shows promise but struggles to keep up with the constantly changing layouts of the UAV networks. To address these issues, SDN offers a centralized control mechanism that adapts to real-time network changes, enhancing routing efficiency and scalability [7].

SDN separates the control plane from the data plane, enabling a centralized controller to maintain a global network view. However existing SDN-based the UAV routing solutions struggle with real-time adaptability in dynamic topologies due to high control overhead, delayed network state updates, and inefficiencies in handling mobility. Our approach overcomes these limitations by integrating SDN with 5G-enabled ground stations, the UAV communication benefits from ultra-low latency, high bandwidth, and edge computing capabilities. This combination ensures seamless data dissemination, minimizes routing overhead, and improves network resilience against failures [7].

In this paper, we propose GeoSDN-UAV, a novel geographic routing protocol to disseminate data from a source node to a target node based on GPSR [8], which integrates SDN and 5G ground stations to improve routing efficiency, scalability, and reliability in the UAV networks. In short, the main contributions are summarized as follows:

- A geo-based routing mechanism designed for dynamic the UAV topologies.
- SDN-based centralized control for real-time routing optimization and reduced overhead.
- Using ultra-low latency 5G communication to improve ground-to-air communication between the UAVs.
- Comprehensive performance evaluations that demonstrate superior packet delivery, reduced delay, and reduced

routing overhead compared to existing protocols.

The remaining sections are structured as follows: Section II reviews related work. Section III details the GeoSDN-UAV architecture and routing mechanism. Section IV outlines the simulation setup and evaluation metrics, followed by a performance evaluation of our approach advantages. Finally, Section V concludes the paper and suggests future research directions.

II. RELATED WORKS

Recent studies have explored the benefits of using the UAVs with SDN technology, which improves the adaptability and stability. In a related study, Berat et al. [9] proposed a new routing algorithm based on the SDN paradigm and exploits central SDN controllers for FANETs. It comprises both proactive and reactive components to improve the performance.

Zhou et al. [10] introduced a FLBD-GPSR, an adaptive geographic routing protocol that divides the network into different influence regions. FLBD-GPSR then uses a fuzzy logic system to dynamically adjust the beacon period based on factors like neighbor density, speed similarity, and position stability. It also implements a dynamic timeout adjustment mechanism for neighbor detection.

Bousbaa et al. [11] proposed a GeoUAVs protocol to deliver data to specific geographical zones. Each the UAV determines its position using GPS, and the Transmission Zone (TZ) is defined as a square area. The source UAV initiates communication by sending a geocast packet, which includes the coordinates of the target zone, a time interval for data validity, and the information content. The packet dissemination process ensures that all the UAVs within the TZ receive the data efficiently by minimizing redundant transmissions and optimizing relay selection based on proximity to the zone's boundaries.

The authors in [12] introduce a Geolocation-Based Hopless Opportunistic Routing protocol (GPHLOR) for the UAV networks that leverages both hopless and location-based routing. It exploits all transmission options, regardless of the number of hops or connection quality. GPHLOR selects relays based on geographic location rather than network topology, making it highly resilient to frequent topology changes. Each node independently computes its forwarding priority based on local location data, minimizing the need for frequent network updates. This reduces the protocol overhead, while ensuring efficient transmission.

Similarly, Sudesh et al. [13] proposed a Link Optimized Cone-Assisted Localization (LoCaL) routing protocol, to improve the connection duration between the UAVs, where the source selects the forwarding node from a given set of neighbors by estimating the parameters of remaining energy, connection duration, and security level. In addition, the utility function is used to mathematically formulate the suggested method, which reduces routing overhead in route discovery by choosing all relay the UAV in the cone-shaped request zone, and improving route stability.

TABLE I
SDN-BASED UAV ROUTING VS. GEOGRAPHIC UAV ROUTING

Feature	SDN-Based UAV Routing	Geographic UAV Routing
Control Mechanism	Centralized (SDN Controller)	Distributed (Position-Based)
Adaptability to Topology Changes	High (Global View)	Moderate (Local Decisions)
Scalability	Limited (Single Controller Bottlenecks) / Improved with Multi-Controller	High (No Central Dependency)
Routing Overhead	Moderate to High	Low (Minimal Overhead)
Latency	Low (Fast Route Computation)	Low to Moderate (GPS Accuracy Dependent)
Reliability	High (Stable with Centralized Control)	Moderate (GPS Errors, Location Updates)
Suitability for Large UAV Swarms	Requires Multi-Controller Setup for Scalability	Well-Suited (Handles Large-Scale Networks)

Research on routing in the UAV networks by leveraging SDN, aims to address the limitations of conventional the UAV networks by improving throughput and enabling virtualization. In this context, our GeoSDN-UAV protocol is mainly based on SDN and 5G-enabled ground stations to improve the UAV network communications. Thus, our protocol leverages the ultra-low latency of 5G communication SDN controllers to ensure proactive routing optimization and minimize packet loss, transmission delay, and network congestion. While SDN enhances global network control, geographic-based routing offers a decentralized alternative. A comparative analysis of these approaches is provided in Table I, outlining their strengths and limitations.

III. OUR GEOSDN-UAV PROTOCOL

In this section we describe our proposed approach, GeoSDN-UAV: A Geographic Unicast Routing Protocol for the UAV Networks Using SDN and 5G-Enabled Ground Stations. This mechanism is based on GPSR which utilizes the geographic positions of UAVs to make forwarding decisions dynamically. GeoSDN-UAV integrates Geographic Unicast communication with SDN-based network management to propose efficient and reliable paths between two nodes of the UAV networks. Our approach enhances routing performance by selecting the optimal path within a defined geographical position while dynamically adjusting routes based on real-time network conditions. By integrating the SDN controller, the system maintains a global view of the network, allowing intelligent routing decisions, while 5G-enabled ground stations provide low-latency communication. In the following, we present this approach in detail. Before that, we introduce a mathematical model to formally describe our proposed system.

Let us define our mathematical model of the swarm as an undirected graph $S = \langle U, E \rangle$, where U denotes a set of UAVs, and E is a set of communication links. Let us define a $U = \{U_1, U_2, \dots, U_n\}$ as a set of UAV nodes moving in a 3D space. Each the UAV U_i for a time t has a position vector in 3D coordinates: $P_i(t) = (x_i, y_i, z_i)$ where x_i, y_i, z_i represent the longitude, latitude, and altitude of the UAV U_i , we assume

that these coordinates are provided by the localization system that provides its real-time geographic like GPS. The velocity of the UAV U_i is given by $V_i(t) = (v_{x_i}, v_{y_i}, v_{z_i})$. The UAV's trajectory over time follows the mobility model that defines a

$$P_i(t + \Delta t) = P_i(t) + V_i(t) + \Delta t \quad (1)$$

where Δt is the time interval between updates.

A link $l \in E$ is denoted by $\langle U_i, U_j \rangle$ and $i \neq j$ directly connecting drone $U_i \in U$ to drone $U_j \in U$, and, in this case, U_i and U_j are said to be neighbors if they are within the communication range of each other. We define $N(U_i)$ as the set of all neighbors of drone U_i .

In GeoSDN-UAV, the packet is forwarded to the neighbor closest to the destination. The distance between a U_i and its neighbor U_j is computed using the 3D Euclidean distance:

$$D(U_i, U_j) = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2} \quad (2)$$

Instead of selecting a neighbor purely based on distance, we incorporate a stability factor:

$$S(U_i, U_j) = \frac{RSSI_{ij}}{\Delta V_{ij}} \quad (3)$$

Where $RSSI_{ij}$ is the Received Signal Strength Indicator between U_i and U_j . ΔV_{ij} is the relative velocity between the UAVs:

$$\Delta V_{ij} = |V_i - V_j| \quad (4)$$

The final link stability weight used for UAV selection is:

$$W_{ij} = \alpha D(U_i, U_j) + \beta S(U_i, U_j) \quad (5)$$

Where α and β are tunable parameters to balance distance with link stability. Then a UAV with minimum weight is selected as the next hop.

The right-hand rule selects the next-hop UAV node based on the minimum angle between the destination vector and the available neighbor links. The angle θ_k for each neighbor U_k is computed as follows:

$$\theta_k = \cos^{-1} \left(\frac{\mathbf{V}_{i,d} \cdot \mathbf{V}_{i,k}}{\|\mathbf{V}_{i,d}\| \|\mathbf{V}_{i,k}\|} \right) \quad (6)$$

where:

- $\mathbf{V}_{i,d} = (x_d - x_i, y_d - y_i, z_d - z_i)$ is the vector from the current U_i to the destination.
- $\mathbf{V}_{i,k} = (x_k - x_i, y_k - y_i, z_k - z_i)$ is the vector from the current U_i to neighbor U_k .
- $\|\mathbf{V}\|$ represents the magnitude (Euclidean norm) of a vector.

We define a path $R(S, D)$ as a sequence of links connecting drones S to D , and R as the set of all possible paths in S . That is, $R(S, D)$ is defined recursively as $R(S, D) = (S, U_i, \dots, U_j, D)$.

We assume that each the UAV U_i periodically transmits its status to the SDN controller through the 5G-enabled ground station. The message includes: unique the UAV identifier ID_i ,

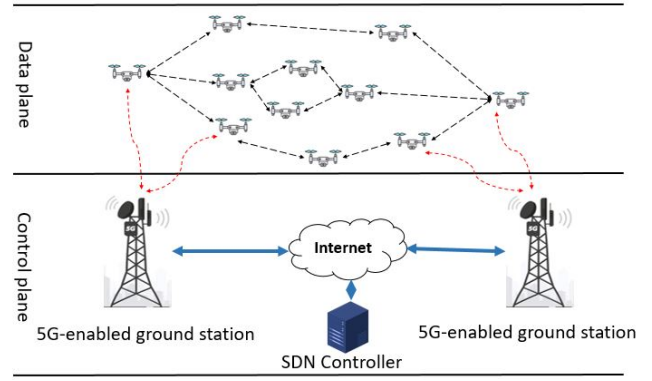


Fig. 1. GeoSDN-UAV Architecture

position (x_i, y_i, z_i) , velocity $V_i(t)$, flight plan, the UAV's direction vector θ_i , and Timestamp of the update T .

$$Status(U_i) = \{ID_i, (x_i, y_i, z_i), (v_{x_i}, v_{y_i}, v_{z_i}), \theta_i, Flight_Plan_i, T\} \quad (7)$$

Each the UAV U_i transmits this information at intervals ΔT to allow the SDN to maintain a global view of the mobility of the UAV and the network topology in real-time. This enables more efficient routing, reduces packet loss, and improves network resilience.

This section outlines the key steps involved in the proposed GeoSDN-UAV system. In the following, we describe the Geographic-Based Forwarding Mechanism using SDN and 5G-enabled Ground Stations. Before that, we describe the architecture of the system, including the role of 5G-enabled ground stations, the SDN controller, and the UAV communication framework.

A. Architecture of the Proposed GeoSDN-UAV System

GeoSDN-UAV framework is a geographic routing-based SDN to improve routing efficiency in the UAV networks. The GeoSDN-UAV architecture as presented in figure 1, is built on three main components: 5G-enabled ground stations, an SDN controller, and a UAVs nodes. Each plays a crucial role in routing performance. In the following, we present components.

One of the key elements of this architecture is the 5G-enabled ground station, which serves as the main communication hub between the UAVs and the SDN controller, where each data exchange must be transmitted over 5G enabled. GeoSDN-UAV selects 5G as a prompt solution to relay, where integration of 5G in the architecture brings significant advantages, including ultra-low latency, high bandwidth, and reliable connectivity. These features allow the UAVs to receive fast SDN updates. In a 5G-enabled ground station, two interfaces are essential for architecture. The first interface connects the UAV to the 5G network, and the second interface links the 5G network to an SDN controller through the Internet.

Secondly part of the GeoSDN-UAV architecture is the SDN controller. It offers a global perspective of the UAV network. It continuously monitors the flow of the control as the positions of UAVs in the swarm, the quality of communication links, and

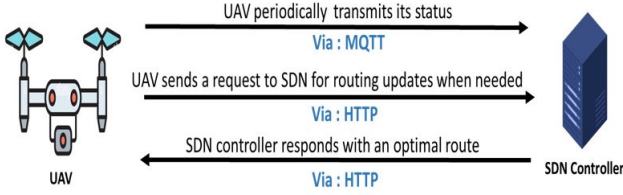


Fig. 2. Communication Flow : UAV & SDN Controller

the overall state of the network. Based on these real-time data transmitted through 5G-enabled ground stations, the controller dynamically adjusts routing paths ensuring messages are sent only to the UAVs within a specific geographic location to ensure the optimal path.

The final component of the architecture is the UAV communication framework, which enables both UAV-to-UAV (U2U) and UAV-to-Ground (U2G) communication. The UAVs communicate directly with one another to relay packets within the geographic position, reducing dependency on the ground station for every transmission. At the same time, the UAV connected to the 5G ground station permanently to receive routing updates and control instructions from the SDN controller. The UAVs continuously update their position using GPS, allowing the SDN controller to adapt routes dynamically.

GeoSDN-UAV architecture uses MQTT (Message Queuing Telemetry Transport) and RESTful API (HTTP) to guarantee the communication flow between the SDN controller and the UAV (represented in figure 2). MQTT is lightweight, adapted to the UAV networks, fast, and ideal for real-time communication, which is used by the UAVs nodes to periodically transmit their status. The UAV uses HTTP to send a request to the SDN for routing updates when needed and the SDN controller uses it to respond with an optimal route to the UAV.

B. Geographic Unicast Routing Mechanism in GeoSDN-UAV

GeoSDN-UAV integrates GPSR, as a geographical unicast routing protocol. The main goal of GPSR is to select the best next-hop the UAV toward the destination using the greedy forwarding strategy. However, when the greedy approach fails due to a local minimum or a network void, the protocol switches to perimeter routing, ensuring reliable packet delivery. Our proposal adapts GPSR for the UAV environments by extending it to 3D space, incorporating link stability metrics, and integrating SDN for real-time routing. In the following, we describe the routing process and how SDN enhances GPSR in the UAV networks which enables dynamic route selection.

In the greedy forwarding strategy, each UAV node forwards packets to the neighbor closest to the destination using the minimum link stability weight. When a UAV U_i receives a packet, it selects the next-hop the UAV U_{next} as

$$U_{next} = \arg \min_{j \in N(i)} W_{i,j} \quad (8)$$

This process ensures that the most stable and closest neighbor is chosen and that packets are sent along the shortest path

Algorithm 1 GeoSDN-UAV : Unicast Routing

Require:

- 1: Source UAV S coordinates (x_s, y_s, z_s)
- 2: Destination UAV D coordinates (x_d, y_d, z_d)
- 3: Neighbor UAV list $N(S)$

Ensure:

- 4: Next-hop UAV for packet forwarding
- 5: **procedure** GPSR_ROUTING
- 6: Get UAV S 's current position from GPS
- 7: Get UAV D 's position from SDN
- 8: Initialize $minWeight \leftarrow \infty$
- 9: Initialize $UAV_{next} \leftarrow NULL$
- 10: **for** each neighbor $j \in N(S)$ **do**
- 11: Compute weighted cost:

$$W_{Sj}$$
- 12: **if** $W_{Sj} < minWeight$ **then**
- 13: $minWeight \leftarrow W_{Sj}$
- 14: $UAV_{next} \leftarrow j$
- 15: **if** $UAV_{next} \neq NULL$ **then**
- 16: Forward packet to UAV_{next} (Greedy Forwarding)
- 17: **else**
- 18: Switch to Perimeter Routing (Right-Hand Rule)

to the destination, reducing packet loss and improving overall reliability. However, when a UAV node reaches a local minimum, it means that it does not have a neighbor with a link stability weight lower $W_{i,j}$ than itself. This condition triggers the switch to perimeter mode, ensuring that the packet is forwarded around the void region until greedy forwarding can resume. Where if no neighbor U_j satisfies $W_{i,j} < W_{i,d}$, then U_i it means is in a local minimum, and perimeter mode is activated. Therefore, packets traverse the planar subgraph using the Right-Hand Rule. The next-hop the UAV U_j is selected based on the smallest angle θ_k between $V_{i,d}$ and $V_{i,k}$:

$$U_j = \arg \min_{\forall k \in N(i)} \theta_k \quad (9)$$

During perimeter traversal, UAVs nodes continuously check if any neighbor U_k satisfies $W_{i,k} < W_{i,d}$. When this happens, the packet switches back to Greedy Forwarding Mode and continues toward the destination. If a UAV node is in perimeter mode for too long, SDN reroutes traffic through more stable UAV node. This mechanism ensures reliable packet delivery in sparse the UAV networks with void regions.

To overcome these limitations about greedy forwarding and perimeter routing, GeoSDN-UAV introduces an SDN-assisted approach where the UAV node first request an optimized route from the SDN controller, ensuring stable next-hop selection. Where before forwarding a packet, UAVs U_i request an optimized route from the SDN controller, enabling predictive routing:

$$Route_Info = SDN_Controller(U_i, D) \quad (10)$$

Algorithm 2 Perimeter Forwarding

Require: Current UAV U_i , Destination D , Neighbor List $N(U_i)$

Ensure: Next-hop UAV for packet forwarding

```

1: procedure PERIMETER_FORWARDING( $U_i, D$ )
2:   Planar_Graph  $\leftarrow$  Construct from  $N(U_i)$ 
3:   Start_Node  $\leftarrow U_i$ 
4:   Current_Node  $\leftarrow U_i$ 
5:   while Packet not reached destination do
6:     Find the next-hop UAV using the Right-Hand Rule
7:     Select neighbor UAV  $U_j$  such that:

$$U_j = \arg \min_{\forall k \in N(Current\_Node)} \theta_k$$

8:     Forward packet to  $U_j$ 
9:     Current_Node  $\leftarrow U_j$ 
10:    if Current_Node returns to Start_Node then
11:      Drop packet (No available path)
12:      Break
13:    if Greedy Forwarding becomes possible then
14:      Switch back to Greedy Forwarding
15:      Break

```

If an optimized path is available, the SDN controller selects the most stable next-hop the UAV based on the link stability weight $W_{i,j}$, which considers both distance and link quality:

$$U_{next} = \arg \min_{j \in N(i)} W_{ij} \quad (11)$$

This approach ensures that UAVs avoid weak or unstable connections, reducing packet loss and disconnection risks.

If no optimized path is available, the UAV falls back to adapted GPSR using greedy forwarding or perimeter routing to navigate in the network. This hybrid SDN-GPSR approach significantly reduces routing failures, improves packet delivery ratio, and enhances network resilience.

Algorithm 3 GeoSDN-UAV Forwarding Algorithm

Require: UAV node U_i , Destination D , SDN Controller SDN

Ensure: Optimized next-hop UAV U_k

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1: procedure GEOSDN-UAV_FORWARDING( $U_i, D, SDN$ )
2:   Route_Info  $\leftarrow$  Request optimal path from  $SDN$ 
3:   if  $SDN$  provides optimized next-hop  $U_k$  then
4:     Forward packet to  $U_k$ 
5:   else
6:     Use (Greedy_Forwarding)

```

IV. EVALUATION PERFORMANCE

In the performance evaluation section, we tested GeoSDN-UAV using OMNeT++ 6.0.3, a reliable simulation tool for modeling communication networks, combined with the INET 4.5.0 framework. We set up a network area of 1.5 km by 1.5 km with a height of 50 meters. The experiments involved 20 to 100 the UAVs. The simulation compared GeoSDN-UAV,

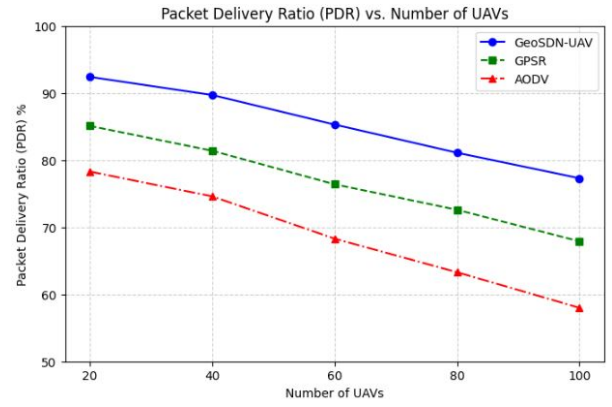


Fig. 3. Packet Delivery Ratio (PDR)

GPSR, and AODV (Ad Hoc On-Demand Distance Vector) [14] under identical conditions over 240 seconds. The evaluation focused on key metrics such as packet delivery ratio (PDR), end-to-end delay, and routing overhead. For a detailed breakdown of all simulation parameters, refer to Table 2, which includes additional

TABLE II
SIMULATION PARAMETERS

Parameters	Values
Network Size	1500 m × 1500 m × 50 m
UAVs Simulated	20, 40, 60, 80, 100
UAV Mobility Model	Random Waypoint Model
Node speed	5 - 20 m/s
Packet Size	512 bytes
Communication Range	250 m
Traffic Type	CBR (Constant Bit Rate)
Routing Protocols	GeoSDN-UAV, GPSR, AODV
Bandwidth	2 Mbps
Evaluation Runs	10

In Figure 3, we compare the packet delivery ratio (PDR) vs. the number of the UAVs. The results show the advantages of integrating an SDN controller with GPSR in 5G enabled the UAV networks. Taking advantage of global network awareness provided by the SDN controller, GeoSDN-UAV achieves superior packet delivery, particularly in scenarios with high mobility of UAVs. As the number of UAVs increases, congestion, interference, and topology changes cause a decline in PDR across all protocols. While GPSR and AODV experience significant drops, GeoSDN-UAV maintains a more stable performance. This is attributed to SDN-assisted route optimization, real-time the UAV status updates, and predictive forwarding decisions. Which helps GeoSDN-UAV to get the best performance.

In Figure 4, we compare the End-to-End Delay (E2E Delay) vs. number of the UAVs. The results show to highlight the effectiveness in terms of optimization packet transmission times of the GeoSDN-UAV protocol. Which achieves lower delays compared to GPSR and AODV. This is primarily due to the proactive route optimization provided by the SDN controller, which selects the most stable next-hop based on real-time

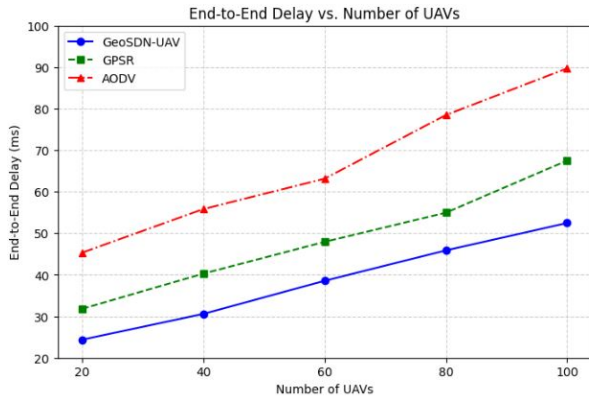


Fig. 4. End-to-End Delay

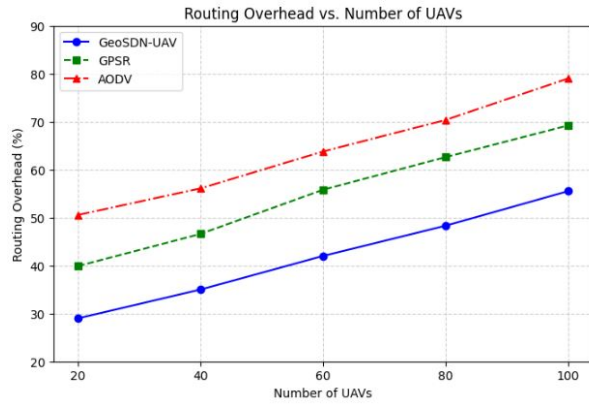


Fig. 5. Routing Overhead

UAV node link quality. At low the UAVs densities (20 UAVs), the delay difference between protocols is smaller, but as the densities scale up, the SDN controller enables GeoSDN-UAV to maintain best performance in delay, proving the impact of SDN-assisted predictive routing and lower latency to request route due to 5G enabled where make fast decision.

In Figure 5, we compare The Routing overhead vs. the number of UAVs. The results demonstrate the efficiency of the GeoSDN-UAV protocol in minimizing control message exchanges, especially in highly dynamic the UAV networks. With lower the UAVs densities (20 UAVs), the difference in overhead between the three protocols is moderate. When the number of UAVs increases, the AODV and GPSR overhead increases dramatically, compared to only 55.5% for the GeoSDN-UAV. This showcases the advantage of SDN-assisted routing, where the controller pre-computes stable paths, reducing unnecessary route requests and retransmissions, thereby significantly lowering routing overhead.

V. CONCLUSION

In this paper, we present a GeoSDN-UAV protocol. A novel geographic unicast routing protocol that is based mainly on SDN and 5G enabled ground stations to enhance the UAV network communication. Our approach builds upon GPSR

and extends it by incorporating SDN-driven routing decisions, real-time topology awareness, and link stability metrics. Our scheme combines ultra-low latency 5G communication and the SDN controller to ensure proactive route optimization, minimizing packet loss, transmission delays, and network congestion. The simulation results show that GeoSDN-UAV significantly improves key network performance such as packet delivery ratio, end-to-end delay, and routing overhead.

In future work, several enhancements can improve GeoSDN-UAV. Firstly, as the UAVs move across different regions, the 5G core network synchronizes status updates among multiple SDN controllers. Secondly, improvements in GeoSDN-UAV to prevent network bottlenecks by monitoring the UAV traffic loads. Finally, we can improve GeoSDN-UAV to predict link failures in advance by using machine learning to guarantee continuous data transmission.

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